

CSC4093/DSC4213 - Neural Networks and Deep Learning

Programming Assignment 02

S/20/346

1. Introduction

This report evaluates the performance of two Convolutional Neural Network (CNN) architectures, VGG16 and ResNet50 for the multi-class classification of tea leaf diseases. The primary objective was to identify the most robust model for accurately distinguishing between three specific conditions: Algal Leaf, Brown Blight, and White Spot.

2. Methodology

Given the strict constraint of the dataset containing only 300 images in total, I relied heavily on transfer learning. Both models utilized base architectures pre-trained on ImageNet. To prevent the networks from simply memorizing the training set and heavily overfitting, I incorporated data augmentation techniques, froze the base feature extraction layers, and trained only the custom classification heads over a span of 10 epochs.

3. Performance Comparison

Both models successfully generalized to the validation data, achieving excellent overall accuracies above 95%. However, diving into the classification metrics revealed a variance in model reliability. VGG16 achieved a higher overall validation accuracy of 97%, outperforming ResNet50's 95%. While ResNet50 showed a faster initial drop in training loss, its validation accuracy fluctuated more across epochs. This indicates that the deeper, more complex residual architecture struggled slightly to stabilize on such a tiny dataset compared to VGG16's more straightforward, sequential base.

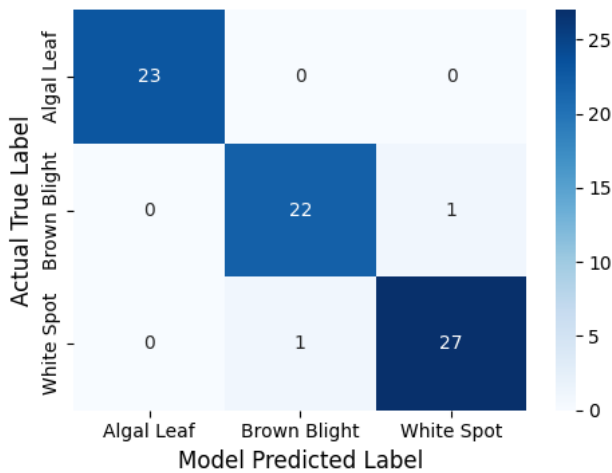
Regarding class-specific confidence, VGG16 demonstrated superior precision. It perfectly isolated the Algal Leaf class (achieving a 1.00 F1-score). Conversely, ResNet50 misclassified several actual Brown Blight images as Algal Leaf, which dropped its precision to 0.92 for that category.

Classification Report

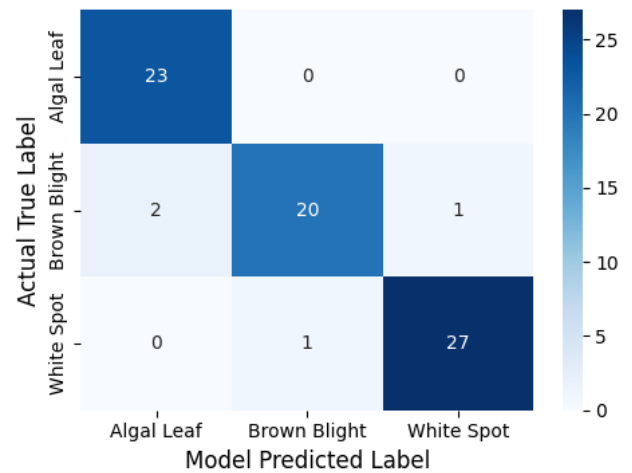
--- VGG16 Classification Report ---				
	precision	recall	f1-score	support
algal leaf	1.00	1.00	1.00	23
brown blight	0.96	0.96	0.96	23
white spot	0.96	0.96	0.96	28
accuracy			0.97	74
macro avg	0.97	0.97	0.97	74
weighted avg	0.97	0.97	0.97	74
--- ResNet50 Classification Report ---				
	precision	recall	f1-score	support
algal leaf	0.92	1.00	0.96	23
brown blight	0.95	0.87	0.91	23
white spot	0.96	0.96	0.96	28
accuracy			0.95	74
macro avg	0.95	0.94	0.94	74
weighted avg	0.95	0.95	0.95	74

Confusion Matrices

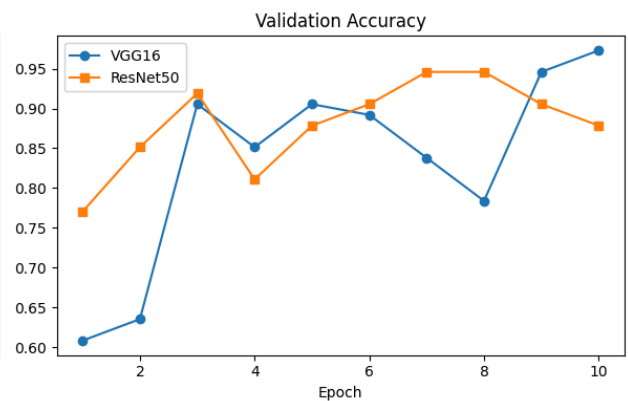
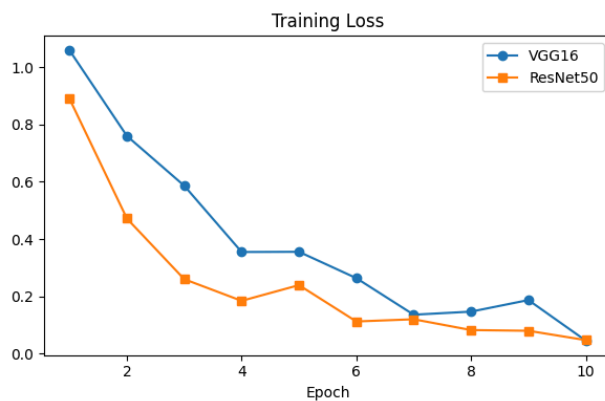
VGG16 Validation Confusion Matrix



ResNet50 Validation Confusion Matrix



Training Loss and Validation Accuracy Plots



4. Conclusion

Finally, I chose ResNet50 as the best model for this project. Even though VGG16 reached a slightly higher peak accuracy (97% vs. ResNet50's 95%), the training plots show that ResNet50 was much more stable. VGG16's accuracy jumped up and down a lot across the epochs, which shows it struggled to learn consistently from this small dataset. On the other hand, ResNet50 improved smoothly and steadily without those extreme spikes. I prioritized having a steady, reliable model over one that just hit a slightly higher peak score. Because of its consistent performance, ResNet50 is the safer and more practical choice.